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Roll No

EC-6003-CBGS

B.E. VI Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Antenna and Wave Propagation

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.
iii) Assume suitable missing data if any.

1. a) Derive expressions for radiation field components (far-fields) due to a current-element. 8
b) Calculate directivity of an antenna producing normalized fields as given ahead. $E_n = \sin \theta$ 6
2. a) What is principle of pattern-multiplication? Explain giving a simple example. 4
b) Plot the radiation pattern of the array shown ahead in Fig.1 and calculate its half power beam width, beam width between first nulls and approximate value of beam area. All the array elements are isotropic-sources having same current amplitude and same phase separated by a distance- $d = \lambda/2$

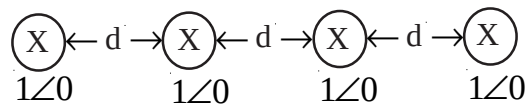


Fig. 1

[2]

3. a) Write short note on Yagi-Vda-Array. 7
b) Discuss characteristic-features and applications of a log-periodic-array. 7
4. a) Derive an expression for radiation pattern of a linear-array of isotropic-sources in terms of a Fourier-Series-elements. 7
b) Discuss Taylor's synthesis-technique in brief. 7
5. a) Define the following: 7
i) MUF and LUF
ii) Virtual height and skip-distance
iii) Critical frequency
b) Discuss space-wave-propagation and sky-wave propagation in brief. 7
6. a) What is Binomial-Array? Discuss in brief. 7
b) Discuss characteristics of rhombic-antenna. 7
7. a) Give classification of antenna-arrays. 7
b) Determine the radiation resistance and power radiated due to a monopole antenna mounted vertically near the earth carrying a current of 10Amp at 3MHz having height of 5-meters. 7
8. Write short notes (any two): 14
i) Tropospheric-propagation
ii) Babine's -principle and complimentary antennas
iii) Two-element-array
iv) Radiation-pattern and directivity

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